

ML Architecture Design Document - Week 5 Actual Implementation

Week 5 - Machine Learning Model Architecture Design

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1. Executive Summary

This document describes the machine learning architecture implemented in Week 5 of the Quantitative Research & Trading project.

The current Week 5 implementation prices chooser options on JPM stock using two complementary approaches:

- Approach 1 (ML + chooser closed form)**: ML models predict 20-day forward realised volatility, and the predicted volatility is passed into the chooser-option closed form.
- Approach 2 (End-to-End)**: ML models directly map chooser contract parameters and market features to chooser prices.

2. Problem Formulation

Approach 1

The first approach estimates the 20-day forward realised volatility and uses that estimate inside the chooser pricing formula.

$$\hat{V}_{\text{chooser}} = C \cdot \left(S, K, T_2, r, q, \hat{\sigma}_{\text{ML}} \right) + P \cdot \left(S, K e^{-r(T_2 - T_1)}, T_2 - T_1, r, q, \hat{\sigma}_{\text{ML}} \right)$$

where $\hat{\sigma}_{\text{ML}}$ is the ML-predicted 20-day forward realised volatility, and $T_1 < T_2$ are the chooser decision time and final maturity.

Target:

$$\sigma_{\text{fwd}, 20d}[t] = \sqrt{252} \cdot \text{std} \left(\ln \frac{S_{t+i}}{S_{t+i-1}} \right)_{i=1}^{20}$$

Approach 2

The second approach predicts chooser price directly from the market feature set and chooser contract variables.

$$\hat{V}_{\text{chooser}} = f_{\theta} \left(S, K, T_1, T_2, r, q, m, \mathbf{x}_{\text{market}} \right)$$

where f_{θ} is a trained ML model and $\mathbf{x}_{\text{market}}$ is the market feature vector.

****Target****: chooser price computed with 20-day historical volatility.

3. Data & Feature Engineering

3.1 Raw Data Sources

Source	Description	Frequency
'yahoo_jpm_2018_2024.csv'	JPM daily close prices	Daily
'fred_DGS10_2018_2024.csv'	US 10-year Treasury yield	Daily
'fred_VIXCLS_2018_2024.csv'	CBOE VIX index	Daily
'jpm_dividends_2018_2024.csv'	JPM dividend payments	Per event
'alphavantage_news_jpm_2018_2024.csv'	News sentiment scores	Per article

3.2 Market Features (Approach 1 - Volatility Dataset)

All features are strictly backward-looking to prevent look-ahead bias.

Item	Value
Dataset rows	1,681 trading days

Feature group	Variables
Historical volatility	'hist_vol_5d', 'hist_vol_20d', 'hist_vol_60d'
Volatility structure	'vol_ratio_5_20', 'vol_ratio_20_60', 'vol_20d_change'
Price momentum	'return_1d', 'return_5d', 'return_20d'
Trend	'price_to_ma_20d', 'price_to_ma_60d'
Risk and sentiment	'vix', 'vix_change_5d', 'vix_ma_ratio', 'vix_jpm_corr_20d', 'r', 'q', 'sentiment_7d', 'sentiment_20d', 'news_count_7d', 'drawdown_20d'

3.3 Features (Approach 2 - Chooser Pricing Dataset)

All market features above, plus chooser-specific parameters.

Item	Value
Dataset rows	17,850 training rows + validation/test rows in the same chooser grid

Feature group	Variables
Market features	all variables from Approach 1

Feature group	Variables
Chooser parameters	`S`, `K`, `moneyness`, `T1`, `T2`

3.4 Target Variables

Approach	Target	Description
1	`fwd_vol_20d`	Annualised realised volatility over the next 20 trading days
2	`chooser_price`	Chooser price using 20-day historical volatility

4. Time-Series Validation Framework

Data is split chronologically, never randomly, to prevent look-ahead bias.

All features are scaled using `RobustScaler` fitted only on the training set.

Split	Date Range	Fraction
Train	2018-03-29 → 2022-11-28	70%
Validation	2022-11-29 → 2023-11-29	15%
Test	2023-11-30 → 2024-12-02	15%

5. Model Architectures

5.1 Approach 1 - ML Volatility Prediction

Random Forest

- `n\_estimators=300`
- `max\_depth=8`
- `min\_samples\_leaf=5`
- Preprocessing: `RobustScaler`
- Input: 21 market features, one row per trading day

XGBoost

- `n\_estimators=500`
- `max\_depth=5`
- `learning\_rate=0.05`
- `subsample=0.8`
- `colsample\_bytree=0.8`
- Preprocessing: `RobustScaler`

LSTM

- Architecture: `LSTM(64) → Dropout(0.2) → LSTM(32) → Dropout(0.2) → Dense(16) → Dense(1)`
- Lookback window: 20 trading days
- Optimizer: Adam
- Loss: MSE
- Early stopping: patience 10
- Input shape: `(batch, 20, 21)`

5.2 Approach 2 - End-to-End Supervised Chooser Pricing

Linear Regression

- Standard ordinary least squares with an intercept term
- Preprocessing: `RobustScaler`
- Input: 26 features covering market variables and chooser parameters

XGBoost

- `n\_estimators=500`
- `max\_depth=6`
- `learning\_rate=0.05`
- `subsample=0.8`
- `colsample\_bytree=0.8`

Neural Network (MLP)

- Architecture: `Dense(128) → ReLU → Dense(64) → ReLU → Dense(32) → ReLU → Dense(1)`
- Optimizer: Adam
- Learning rate: 0.001
- Early stopping via internal validation
- Maximum iterations: 500

6. Performance Summary (Test Set)

6.1 Approach 1 - Volatility Prediction

Model	Vol MAE	Vol RMSE	Chooser MAE	Chooser RMSE
RandomForest	0.07052	0.10154	9.8702	15.8615
XGBoost	0.07394	0.09706	9.8636	15.2889
LSTM	0.07527	0.10495	10.4531	15.3666

Vol MAE/RMSE are measured in annualised volatility units. Chooser MAE/RMSE are measured in USD.

6.2 Approach 2 - End-to-End Chooser Pricing

Model	MAE (\$)	RMSE (\$)	R ²
LinearRegression	5.5223	8.1196	0.7526
XGBoost	8.6511	12.5134	0.4125
NeuralNetwork	3.5388	5.3121	0.8941

The target is the chooser price based on 20-day historical volatility.

The evaluation grid covers two decision times, three maturities, and three moneyness levels.

7. Limitations & Recommended Next Steps

Current Limitations

1. No market-implied volatility is used. The targets are derived from historical or closed-form prices rather than live option quotes.
2. The chooser grid is simplified to two decision times, three maturities, and three moneyness levels.
3. The feature set is static and does not include real-time microstructure information such as bid-ask spread or trading volume.
4. The second approach mainly learns a synthetic chooser surface, while real market prices may still show skew or smile effects.

Recommended Next Steps (Week 6+)

1. Incorporate implied volatility data to make the targets closer to market pricing.
2. Add Greeks such as delta, gamma, and vega as engineered features.
3. Introduce Bayesian hyperparameter optimisation.
4. Extend the LSTM to forecast volatility over multiple future steps.
5. Build an ensemble that combines the two approaches.
6. Evaluate the methods on 2025 and later data for a cleaner out-of-sample test.

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